

Mapping Unpaid Care and Economic Activity Across England and Wales

A Multi-Scale Visual Analytics Study, 2011–2021

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A summary extract: abstract, introduction and conclusion.

Full report, methodology and pipeline: github.com/EbaAndBeans/Mapping-Unpaid-Care-E-W

Interactive Tableau workbook: public.tableau.com/views/ru22097_Tableau_Workbook

Abstract

Around five million people across England and Wales provide unpaid care worth £162 billion each year, yet between 2011 and 2021 the overall share of residents providing any care fell from 11.4% to 9.0% while those providing 20 or more hours per week rose from 1.5% to 1.9%. This project was built for public health analysts and care commissioners who need to understand where that divergence between falling prevalence and rising intensity has landed hardest. A four-dashboard Tableau workbook examines census data at two geographic scales, combining dimensionality reduction, neighbourhood clustering and Bayesian residual modelling to connect national patterns to neighbourhood-level detail.

Every one of the twelve Local Authorities with the largest rises in intensive care, including Hartlepool and Eastbourne, also saw economic inactivity increase over the same decade. Local Authority averages masked wide neighbourhood variation, with Newcastle upon Tyne spanning 3.63 percentage points across its neighbourhoods despite sitting close to the national mean of 4.6%. The highest intensive-care rate of 6.5% appeared not in the oldest retirement-heavy areas but in a health-limited cluster, suggesting that intensive care aligned more closely with long-term sickness and disability than with age alone. The most ethnically diverse areas consistently recorded the lowest care rates, which challenges common assumptions about informal family-based care. In East Lindsey, three neighbourhoods still exceeded model expectations by 2.4 to 3.1 percentage points even after accounting for their already old and inactive populations. A formative evaluation showed that usability depended on visible comparison rules rather than analytical complexity, a finding that generalises well beyond this particular topic.

Introduction

Every day, around five million people across England and Wales quietly provide unpaid support to someone living with a long-term illness, disability or age-related need. This support, which can mean helping someone wash, take medication, move around their home or simply cope, has been valued at £162 billion per year, a 29% increase since 2011 (Centre for Care and Carers UK, 2023). That figure is roughly comparable to the entire annual NHS budget. Yet unpaid care generates no wages, no employment records and no market transactions. It is largely invisible to conventional economic measurement, and the only data source that captures where it falls heaviest at neighbourhood scale is the census.

Between 2011 and 2021, the national numbers told a contradictory story. The overall proportion of residents providing any unpaid care fell from 11.4% to 9.0%, yet the proportion providing 20 to 49 hours per week rose from 1.5% to 1.9%, and those providing 50 or more hours rose slightly from 2.7% to 2.8% (ONS, 2023). Fewer people appear to be caring overall, but those who are caring are doing considerably more of it. Brimblecombe et al. (2018) show that this kind of intensive caring tends to concentrate among those least able to combine it with paid work, which makes understanding where that burden falls a matter of direct relevance to health,

employment and social care policy. This project was built to map that divergence and find the places where it has appeared most sharply.

A national statistic alone cannot answer the questions that matter most to the people who plan and commission care services. How did care rates change across Local Authorities between 2011 and 2021? What neighbourhood-level variation lies hidden beneath those averages? And which areas show care intensity that is unexpectedly high relative to their demographic and economic context, rather than simply high in absolute terms? Answering these questions means moving between scales and types of analysis in a way that a single map or statistical bulletin cannot support. Visual analytics is designed to do exactly that. Keim et al. (2010) describe it as the combination of automated computational analysis with interactive visualisation, allowing humans to detect patterns at a scale neither method could reach alone. The dashboards follow Shneiderman's (1996) principle of overview first, then zoom and filter, then details on demand. Users start with a national picture of Local Authority change, select an area they want to inspect and work progressively deeper into neighbourhood patterns, similarity between areas and model-based comparisons. Milligan and Wiles (2010) argue that care is itself inherently geographical, shaped by the places in which it is given and received, which makes this kind of spatial analysis a natural fit for the subject.

In this project, unpaid care refers to unpaid help or support provided to someone because of a long-term physical or mental health condition, illness or age-related need, as recorded in the census. It excludes care provided through paid employment and does not cover general childcare or voluntary work unless directly linked to illness, disability or ageing. Two geographic scales are used throughout. Local Authorities provide the unit for 2011 to 2021 comparison, since they are familiar planning units and large enough for change over time to be meaningful. Middle Layer Super Output Areas, referred to throughout as MSOAs, provide the neighbourhood layer for 2021 and reveal the variation that Local Authority averages can conceal. A full MSOA-level comparison between 2011 and 2021 was avoided because MSOAs are small enough that differences in census question wording or population denominators would make change look larger or more precise than it really is. Economic activity is analysed alongside unpaid care as a descriptive area-level variable rather than a cause. Because the data is aggregated by area, the analysis cannot identify the circumstances of individual residents.

The intended users are social care commissioners, public health analysts, Local Authority planning teams and academic researchers. Their needs differ in important ways. A commissioner needs to identify neighbourhoods where intensive care burden is unusually high relative to the area's demographic profile rather than simply high in absolute terms, because high intensive care rates in an older and sicker area are broadly expected, whereas equally high rates in a younger and healthier area are more surprising and may need closer attention. Existing tools such as ONS Census Maps, Nomis downloads and published statistical bulletins handle single-variable, single-scale questions well but do not allow users to move freely between temporal change, neighbourhood distributions, similarity-based area groupings and model-derived comparisons within one system. The dashboard is designed to fill that gap. Two research questions structure the project: how unpaid-care prevalence, intensity and economic activity changed across Local Authorities between 2011 and 2021, and where that contrast was sharpest; and what neighbourhood-level variation is concealed beneath Local Authority averages in 2021, and which MSOAs show unusually high or low intensive care rates relative to their demographic and economic context. To answer the second, PCA, t-SNE and UMAP reveal neighbourhood types and cluster structure, and a Bayesian Ridge regression model estimates expected intensive care from area context to flag neighbourhoods where observed care departs from prediction.

Conclusion

Between 2011 and 2021, the headline story for unpaid care in England and Wales was a falling one. Fewer people were recorded as carers overall, but beneath that decline intensive caring quietly grew. The Isles of Scilly, Hartlepool and Eastbourne all saw 20 or more hours care rise by over a percentage point, and every one of the twelve highest-change Local Authorities also saw economic inactivity increase over the same decade. The divergence was not random: in the highest-change authorities, rising intensive care coincided with rising economic inactivity.

Local Authority averages, however, obscured as much as they revealed. Newcastle upon Tyne had a 20+ hours care rate of 4.4%, fractionally below the England and Wales mean of 4.6%, yet the spread across its neighbourhoods was 3.63 percentage points, the widest of any Local Authority. Kensington and Chelsea (3.25pp) and Brighton and Hove (2.97pp) showed the same pattern, where the average looked unremarkable but the distribution beneath it did not. More strikingly, the highest intensive care rate of 6.5% appeared not in the oldest retirement-heavy areas, which had by far the largest share of residents aged 65+, but in the health-limited cluster where long-term sickness and disability were more than twice the national average. Planning services around age profiles alone would miss the communities carrying the greatest burden.

A further assumption the data challenged was that more ethnically diverse areas carry heavier informal care burdens, as family-based care cultures are sometimes assumed to fill gaps left by formal provision. The three clusters with the highest ethnic diversity, covering diverse younger urban, mixed working-age and student-heavy areas, recorded the three lowest intensive care rates. The most diverse cluster reached only 4.2% despite having the highest ethnic mix of any group, while the least diverse cluster had a higher intensive care rate despite a sevenfold gap between them in the ethnic diversity index. This does not explain why care rates are lower in more diverse areas, but it does suggest the assumption deserves scrutiny rather than acceptance.

The modelling layer then separated areas with high care in absolute terms from those where care exceeded what their local profile would predict. East Lindsey was the clearest case, with three of its neighbourhoods already among the oldest and most economically inactive in England and Wales, yet still recording 20+ hours care rates 2.4 to 3.1 percentage points above model expectation. Higher-than-expected residuals appeared more concentrated in parts of Wales and the Midlands, while lower-than-expected cases were scattered without any clear geographic pattern. Because the residual flags isolate the highest and lowest 5%, these cases are best read as exceptional neighbourhoods for closer inspection rather than evidence of a widespread national pattern.

The central visualisation lesson was that analytical complexity and usability are not the same thing. The Bayesian residual dashboard was technically the most demanding component yet was the easiest for participants to use, because its comparison rule was clearly stated and observed care was visibly either above, below or close to model expectation. The projection dashboard was the hardest, not because it was more complex, but because it asked users to read distance and cluster profiles without enough guidance on what those things meant. Making the comparison rule explicit matters more than methodological sophistication.

The workbook remains an exploratory tool rather than a prescriptive guide. All findings describe neighbourhood patterns rather than the circumstances of individual carers, and the cluster labels are descriptive area types rather than fixed categories. The most important next steps are validation with professional users, stronger benchmarks that compare each Local Authority with similar places, and a residual model that accounts for local service provision, because a community's care burden cannot be fully understood without knowing what formal alternatives exist nearby.